RGB-D NOVEL-VIEW RECONSTRUCTION

Geometry-Supervised 3D Gaussian Splatting

Registered depth guides surface distance, local orientation, and removal of Gaussians whose centers are repeatedly contradicted in measured free space.

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FULL GEOMETRY VERSUS RGB LOSS ONLY · SAME VIEWS, RGB-D INITIALIZATION, AND 10K SCHEDULE

−34.7%
Depth RMSE
0.453 m → 0.296 m

−25.5%
Mean angular normal error
45.2° → 33.7°

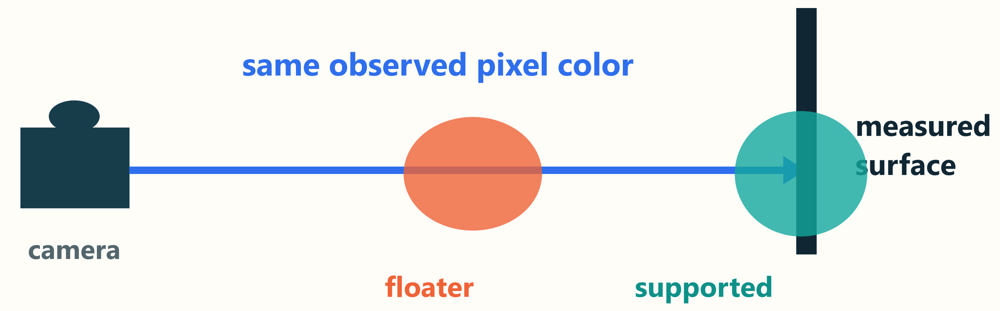
−29.6%
Unsupported centers
17.79% → 12.53%

Held-out RGB quality (PSNR, higher is better) changes by +0.09 dB · 19.183 → 19.275 dB

01 The problem

3DGS represents a scene with many translucent colored 3D ellipsoids called Gaussians. RGB supervision can place the correct color at several depths along one camera ray. A misplaced Gaussian may fit the training image and appear as a floater from another viewpoint.

ONE TRAINING RAY



Sensor depth identifies the measured surface and exposes the nearer Gaussian as unsupported free-space geometry.

Held-out evaluation metric: percentage of centers with valid depth checks in at least two held-out views that fall clearly in front of the measured surface in at least half of those views. Lower is better.

INPUTS AND OUTPUTS

Training input
- RGB image
- Registered sensor depth
- Calibrated camera pose

Learned output
- 3D Gaussian scene
- Novel-view RGB
- Novel-view depth proxy in metres

RESEARCH QUESTION

Can registered RGB-D supervision improve 3DGS geometry while preserving held-out RGB quality?

MAIN EVALUATION

3	3	10k
TUM trajectories	random seeds	iterations

Each trajectory uses 84 training views and 12 fixed held-out views at 320×240. All main variants share the RGB-D initialization and train-test split. "RGB loss only" uses depth for initialization, then trains without depth or normal supervision. Held-out RGB and depth are excluded from training and used for evaluation.

02 How geometry enters training

The standard RGB path stays active. A second path derives depth and normal losses from registered sensor depth, while a third path removes unsupported centers.

THREE ALIGNED PATHS

Current Gaussians + camera pose →	3DGS renderer →	Rendered RGB + training RGB → RGB loss
Renderer depth output + sensor depth →	Metric depth and local normals →	Depth + normal losses
Gaussian centers + sensor depth →	Cross-view support check →	Remove Gaussians with unsupported centers

The RGB, depth, and normal losses update Gaussian parameters. At pruning steps, the method deletes Gaussians whose centers repeatedly conflict with sensor depth.

01 Metric depth consistency

Goal: place rendered surfaces at measured distances.

$$\mathcal{L}_{\mathrm{depth}} = \mathrm{mean}\left|\log \hat{D} - \log D\right|$$

$\hat{D}$: rendered metric depth · D: registered sensor depth · mean over valid positive depths

Convert the renderer's opacity-weighted inverse-depth output to a depth estimate in metres. Compare rendered and sensor depth with a log-depth loss, which treats equal relative errors similarly across distance.

02 Normal consistency

Goal: stabilize local surface orientation.

$$\mathcal{L}_{\mathrm{normal}} = \mathrm{mean}(1 - |\hat{\mathbf{n}}^T\mathbf{n}|)$$

$\hat{\mathbf{n}}$: rendered-depth unit normal · n: sensor-depth unit normal · mean over valid pairs

Back-project rendered and sensor depth, compute both normal maps with the same local operator, then align their directions. The absolute dot product treats a normal and its reversed direction as equivalent.

03 Cross-view free-space pruning

Goal: remove Gaussians whose centers sit in measured free space.

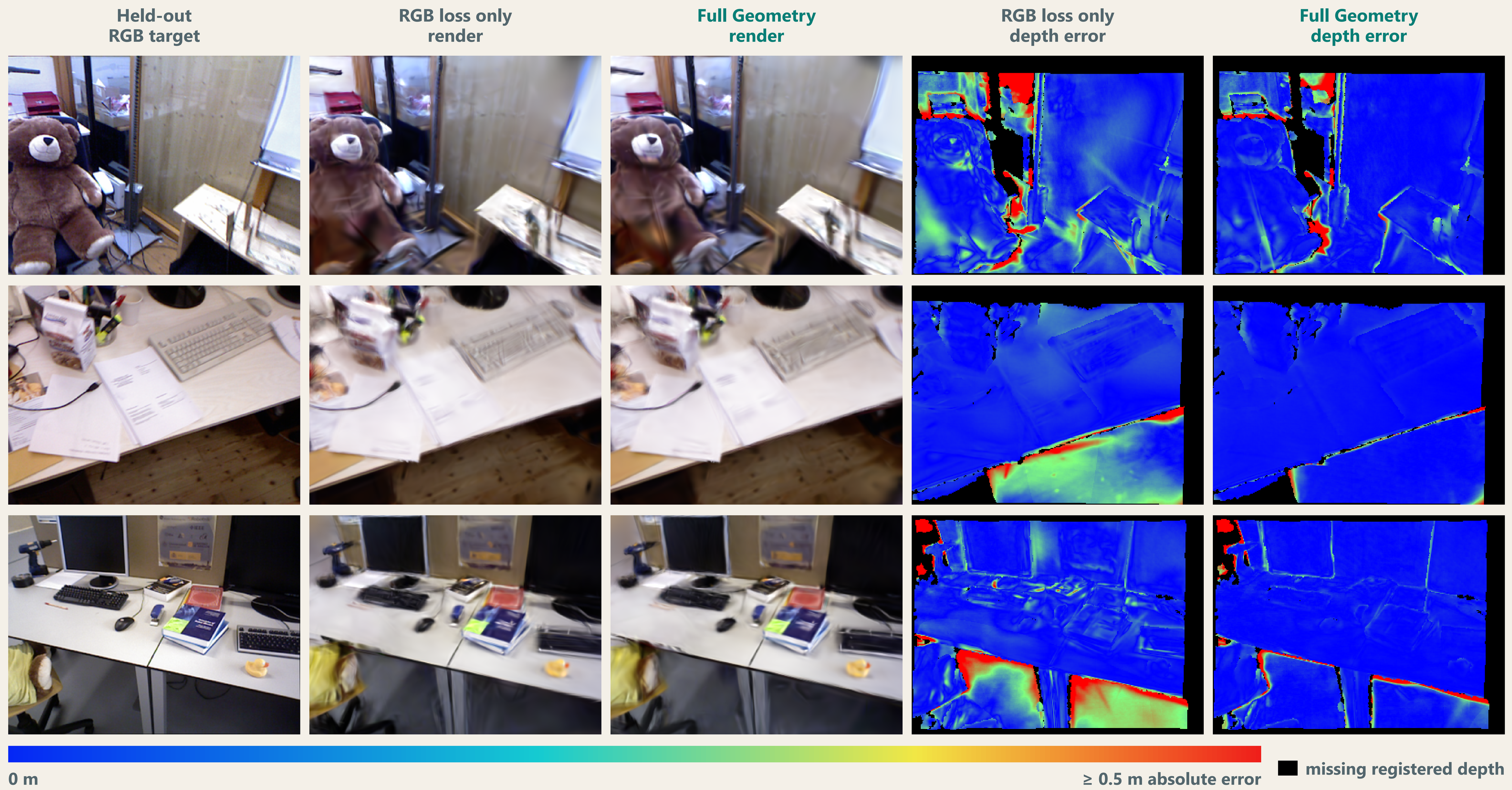
$$z_g < D - \max(0.07\,\mathrm{m},\ 0.05D) \Rightarrow \text{free-space contradiction}$$

z_g: Gaussian center depth · D: registered sensor depth

Project each Gaussian center into several training depth maps. A contradiction places the center farther in front of the measured surface than the fixed 7 cm or 5% margin. Remove a Gaussian after at least four valid checks when 60% or more are contradictions.

03 Held-out views show the change

Full Geometry combines depth, normals, and pruning. These illustrative 10k views use seed 0; the aggregate results below use all 12 held-out views per trajectory.



Visible pattern: In fr1/room, error across the bear torso and nearby floor shifts toward blue, while this view's PSNR changes by −0.11 dB. In fr1/desk, the broad floor band below the desk edge turns blue. In fr3/office, error below the tabletop recedes while thin object edges stay red. Against their held-out targets, the desk and office Geometry renders gain +0.35 and +0.49 dB PSNR.

Add one component at a time

10k macro means · 3 trajectories × 3 seeds

RGB LOSS ONLY → RGB + DEPTH	RGB + DEPTH → RGB + DEPTH + NORMALS	DEPTH + NORMALS → FULL GEOMETRY
Depth RMSE	**Mean angular error**	**Unsupported centers**
−29.6%	**−16.6%**	**−16.3%**
0.453 → 0.319 m	40.9° → 34.1°	14.97% → 12.53%

Full Geometry versus RGB loss only · Depth RMSE falls on **3 / 3** trajectory means · PSNR change **+0.09 dB**

DEPTH RMSE BY TRAJECTORY · RGB LOSS ONLY → FULL GEOMETRY

fr1/room	fr1/desk	fr3/office
0.376 → 0.235 m · **−37.5%**	0.166 → 0.111 m · **−33.2%**	0.818 → 0.542 m · **−33.7%**

COMPLETE-SYSTEM COMPARISON

Same TUM protocol. **DN-Splatter leads:** depth RMSE, 0.199 versus 0.296 m; unsupported centers, 11.21% versus 12.53%. **Geometry leads:** PSNR, 19.275 versus 18.753 dB; macro normal error, 33.72° versus 33.90°; Gaussians, 64k versus 180k.

TAKE-HOME

Depth constrains distance, normals constrain orientation, and cross-view pruning removes Gaussians whose centers are repeatedly contradicted. Together they reduce all three reported geometry errors while held-out RGB PSNR changes slightly.

CODE AND REFERENCES

github.com/alasleimi/geometry-supervised-3dgs

Kerbl et al., 3D Gaussian Splatting, TOG 2023 · Sturm et al., TUM RGB-D Benchmark, IROS 2012 · Turkulainen et al., DN-Splatter, WACV 2025